Example of Pascal's Identity for n=5, k=2

$$\binom{5}{2} = \binom{4}{1} + \binom{4}{2}$$

Let X = {A, B, C, D, E}

S = 2-combinations of X

= {those containing E} U {those without E}

= S_1 U S_2

S_1 = { {A,E}, {B,E}, {C,E}, {D,E} }

i.e., 1-combinations from {A,B,C,D}

$|S_1| = \binom{4}{1}$

S_2 = {{A,B}, {A,C}, {A,D}, {B,C}, {B,D}, {C,D}}

i.e., 2-combinations from {A, B, C, D}

$|S_2| = \binom{4}{2}$

$$\binom{5}{2} = |S| = |S_1| + |S_2| = \binom{4}{1} + \binom{4}{2}$$